



PhenoRx

Pharmacogenomic Adverse Drug Reaction Risk Assessment

A technical and methodological overview of a clinical decision support prototype that combines a machine-learning classifier, pharmacogenomic rule engines, and a large language model to estimate adverse drug reaction risk before a medicine is administered.

Written for medical students and clinicians with no machine-learning background. It explains what the system does, how each component works, and — importantly — how well it actually performs.

This is a teaching and research prototype, not a clinical tool.

It is trained on synthetic data, has never been externally validated, and its measured accuracy is below that of a trivial baseline. It must not be used to make decisions about any real patient. Section 6 explains the evidence for this statement in full.

Document	Technical & Methodological Overview
System version	2.1.0
Model artifact	<code>models/adr_model.pkl</code> (2.5 MB)
Audience	Medical students, clinical pharmacology trainees

Contents

- 1 The clinical problem** Why ADR prediction matters

- 2 What the system does** The clinical workflow, page by page

- 3 Architecture** Three independent layers

- 4 Layer 1 — The machine-learning model** Algorithm, features, training

- 5 Layer 2 — Pharmacogenomic rule engines** CYP, transporters, HLA

- 6 Model performance — the honest numbers** What it scores, and why

- 7 Layer 3 — The AI narrative** Large language model integration

- 8 Safety, limitations and governance** What would be needed clinically

- 9 Glossary** Machine-learning terms in clinical language

How to read this document

Sections 1–3 describe the system and need no technical background. Sections 4–7 go into the methods. **Section 6 is the most important section** — it reports the measured performance, separates the two different questions the model is asked, and derives the mathematical ceiling on how well either can be answered. If you read only one part, read that one: it is the part that teaches you how to interrogate any clinical AI tool you are shown in future.

1 The clinical problem

An adverse drug reaction (ADR) is harm caused by a medicine given at a normal dose for a normal indication. ADRs are a substantial cause of hospital admission and inpatient morbidity, and a meaningful proportion are considered preventable.

The clinical difficulty is that risk is not uniform. Two patients given the same drug at the same dose can have very different outcomes, because risk depends on the interaction of:

- **Organ function** — chiefly renal clearance (eGFR) and hepatic metabolism, which determine how quickly a drug is eliminated.
- **Genetics** — inherited variation in metabolising enzymes and immune receptors, which can multiply exposure or predispose to severe immune reactions.
- **Co-medication** — competing substrates, enzyme inhibition or induction, and additive toxicity across a regimen.
- **Comorbidity and physiology** — age, body composition, cardiac disease, electrolyte state.

A clinician integrates these mentally. The premise of this project is that the same integration can be performed computationally, before the first dose, to flag patients who warrant a different agent, a lower dose, or closer monitoring.

The premise is reasonable; the execution here is a prototype

The idea above is sound and is an active area of real research. This particular implementation does not achieve it — see Section 6. The value of this document is in showing how such a system is constructed and how to evaluate whether it works.

Why "phenotype" plus "Rx"

The name reflects the intended approach: combine the patient's observable clinical phenotype (laboratory values, comorbidities, physiology) with their pharmacogenomic phenotype (metaboliser status) to individualise the prescription.

2 What the system does

The application is a web tool organised around a single clinical assessment. A user works through five screens in sequence.

1 • Patient details

Demographics: age, sex, ethnicity, height and weight. Body mass index is calculated automatically, and age and sex select the laboratory reference intervals used on every subsequent screen.

2 • Clinical assessment

The main data-entry form: renal and hepatic panels, full blood count and differential, vital signs, comorbidities, the drug regimen, and pharmacogenomic results. Values are validated against the same numeric bounds the server enforces, so an implausible entry is flagged as it is typed.

~60 fields, fillable in any order; a demo patient can populate all of them

3 • Report

The model's numerical output and the AI-generated narrative are presented side by side: overall risk percentage, risk tier, predicted reaction classes, contributing factors, and pharmacogenomic profile. Exportable as PDF.

4 • Interactions and dosing

Pairwise interaction checking across the regimen with severity classification, plus renal-, hepatic- and CYP-adjusted dose suggestions with monitoring intervals.

5 • Counselling

A six-step structured medication education workflow, including teach-back comprehension scoring and signed acknowledgement from both patient and counsellor.

Risk tiers

The model returns a probability, which is mapped to one of four tiers. The thresholds are fixed constants in the code, not learned from data:

Risk tier thresholds, from `services/ml_service.py`

TIER	OVERALL ADR RISK	INTENDED INTERPRETATION
Critical	≥ 75%	Consider an alternative agent; specialist input
High	≥ 50%	Dose reduction and active monitoring
Moderate	≥ 25%	Standard dosing with scheduled review
Low	< 25%	Routine care

A point worth noticing

These cut-points are arbitrary round numbers. In a validated clinical tool, thresholds would be derived from the data — chosen to achieve a target sensitivity or to optimise net clinical benefit, then tested on a separate cohort. Fixed round numbers are a reasonable placeholder in a prototype, but they are an assumption, not a finding.

3 Architecture

Three independent layers contribute to the output. They are worth separating clearly, because they have very different reliability characteristics.

LAYER	WHAT IT IS	HOW IT CAN FAIL
1. ML classifier	A gradient-boosting model trained on 50,000 synthetic patient records	Silently wrong. Produces a confident-looking number whether or not it is correct, and cannot signal its own uncertainty. See Section 6 for how well it performs.
2. PGx rule engines	Deterministic rules encoding published gene–drug relationships	Transparent. Rules are explicit and auditable; failure means a rule is missing or out of date, which is inspectable.
3. LLM narrative	A large language model that writes the clinical prose	Fluent but unreliable. Can produce plausible, well-written statements that are incorrect. Degrades to fixed rule-based text when unavailable.

Technology stack

Backend	Python 3.13, Flask (application factory with blueprints)
Machine learning	scikit-learn — <code>HistGradientBoostingClassifier</code>
Frontend	React 19 with Vite, Tailwind CSS, single-page application
Language model	NVIDIA NIM (Nemotron), OpenAI-compatible API
Serving	Gunicorn, one worker with four threads, containerised
Persistence	In-memory only — no database

Why the three layers are kept separate

Separation means a failure in one layer does not silently corrupt the others. If the language model is unreachable, the system still returns the model's number and the pharmacogenomic findings, with deterministic text in place of the narrative. The converse also holds: a fluent AI narrative does not make the underlying risk number any more trustworthy.

4 Layer 1 — The machine-learning model

The algorithm

The model is a **histogram-based gradient boosting classifier** (`HistGradientBoostingClassifier` from scikit-learn). In clinical terms, it can be understood as follows.

A single *decision tree* is a flowchart of yes/no questions — "is eGFR below 30?", "is the patient over 70?" — ending in a prediction. One tree alone is crude. *Gradient boosting* builds hundreds of trees in sequence, where each new tree is trained specifically on the cases the existing trees got wrong. The final prediction is the accumulated vote of all the trees.

This family of model is a sensible default for tabular clinical data: it handles mixed numeric and categorical variables, tolerates missing values natively, and captures interactions between variables without being told to look for them.

Training hyperparameters, from `config/ml_config.yaml` defaults

PARAMETER	VALUE	MEANING
<code>max_iter</code>	400	Maximum number of boosting trees
<code>max_depth</code>	18	Maximum questions deep any one tree may go
<code>learning_rate</code>	0.07	How much each tree is allowed to correct its predecessors
<code>l2_regularization</code>	0.3	Penalty discouraging over-confident fitting
<code>random_state</code>	42	Fixed seed, so training is reproducible
train/test split	80 / 20	A fifth of data held back to test on

Input features

The model consumes **29 variables** per patient:

DOMAIN	VARIABLES
Demographics	age, sex, ethnicity, weight, height, BMI
Renal	creatinine, eGFR
Hepatic	AST/ALT, bilirubin, albumin
Haematology	haemoglobin, white cell count, platelet count
Vital signs	systolic and diastolic blood pressure, heart rate, temperature
Comorbidity	diabetes, liver disease, chronic kidney disease, cardiac disease, hypertension
Drug exposure	index drug dose, concomitant drug count, polypharmacy flag, days since start
Pharmacogenomic	CYP2C9 and CYP2D6 metaboliser status

Note the mismatch between form and model

The assessment form collects roughly 60 fields, including an extensive pharmacogenomic panel (six CYP enzymes, three transporters, three HLA alleles). The model itself uses only the 29 variables above, of which just **two** are pharmacogenomic. The remaining PGx data is used by the rule engines in Layer 2, not by the model. This is worth knowing: entering a full genomic panel does not make the risk *number* more informed than entering CYP2C9 and CYP2D6 alone.

Training data

Records	50,000
Origin	Synthetic — generated by <code>scripts/data_generator.py</code>
Outcome classes	20 distinct ADR types, including "No ADR"
Split	40,000 training / 10,000 held-out test
Commonest class	"No ADR", 22.3% of records
Rarest class	265 records

The size of the training set was itself the main defect

Until recently the model was trained on a **200-row** file spanning **60** outcome classes — roughly three examples per class. The data generator defaults to 200,000 records; it had been run with 200. Regenerating the cohort at 50,000 records reduced the class count to 20, raised the rarest class from 3 examples to 265, and is the single change responsible for most of the improvement reported in Section 6.

The data is synthetic, and this is the root limitation

No real patient contributed to this model. The records were produced by a script that encodes assumptions about how clinical variables relate to adverse outcomes. A model trained this way can only learn the rules its generator already contained — it cannot discover anything about real physiology, and good performance on such data would demonstrate nothing about real-world accuracy.

5 Layer 2 — Pharmacogenomic rule engines

Separate from the model, four deterministic engines encode published gene–drug relationships. These are ordinary conditional logic, not learned from data, which makes them transparent and auditable.

ENGINE	SCOPE
CYP engine	Cytochrome P450 metaboliser phenotypes. Poor metabolisers accumulate an active drug; ultra-rapid metabolisers may fail to respond, or over-produce an active metabolite from a prodrug.
Transporter engine	Membrane transporter variants affecting drug distribution and hepatic uptake — the mechanism underlying, for example, statin-associated muscle symptoms.
HLA engine	Human leukocyte antigen alleles associated with severe immune-mediated hypersensitivity, the class of reaction where pre-prescription genotyping has the clearest established clinical value.
Non-CYP engine	Other metabolising enzymes outside the P450 system.

Why this layer is more trustworthy than the model

A rule stating a relationship between a genotype and a drug can be traced to a published source, reviewed by a pharmacologist, and corrected when guidance changes. Its behaviour is identical every time and can be reasoned about without running it. The machine-learning model has none of these properties: it is a numerical function fitted to data, and its reasoning cannot be inspected directly.

The generalisable lesson

When you encounter a clinical AI product, ask which parts are learned and which are encoded rules. Rules can be checked against guidelines. Learned components can only be trusted to the extent that they have been validated on data resembling your patients.

6 Model performance — the honest numbers

This section reports what the model actually scores, on a held-out fifth of the data it never saw during training. Two different questions are measured separately, because the system answers both and they are not equally difficult.

6.1 Two questions, two very different scores

Held-out test set, 10,000 records. Measured from the shipped artifact.

QUESTION	ACCURACY	BASELINE	CEILING	FIELD
Will this patient have an ADR? Binary	85.35%	77.71%	88.45%	<code>overall_adr_risk</code>
Which ADR will it be? 20-class	52.72%	22.29%	58.27%	<code>predicted_adr_type</code>

The **baseline** is the score from a trivial rule that ignores every input and always answers the commonest class. The **ceiling** is derived in §6.3 — it is the best score any model could possibly achieve on this data, however good.

6.2 The headline number

85.35% ACCURACY, ADR YES/NO Baseline 77.71%	0.930 ROC-AUC 0.5 = coin toss	89.3% RECALL ADRs correctly flagged	91.5% PRECISION Flags that were right
--	--	--	--

For the question the interface actually foregrounds — a single risk percentage — the model discriminates well. An ROC-AUC of 0.93 means that given one patient who went on to have an ADR and one who did not, the model assigns the higher risk to the right one about 93% of the time. Recall of 89.3% means it flags roughly nine in ten of the reactions present.

Predicting *which* reaction is much harder, by construction

At 52.72% the multiclass score looks weak next to the binary one, and it is — but §6.3 shows that 58.27% is the arithmetic maximum. The model reaches 90% of what is attainable. The remaining error is not a modelling failure; it is irreducible uncertainty written into how the data was made.

6.3 Why there is a ceiling at all

The synthetic data generator assigns outcomes with a cascade of clinical rules. The *branch* is deterministic — hepatic impairment routes to liver reactions, renal impairment to kidney reactions — but **which** reaction within that branch is then drawn at random with fixed probabilities:

Outcome assignment, from `scripts/data_generator.py`

BRANCH CONDITION	POSSIBLE OUTCOMES AND THEIR PROBABILITIES	BEST POSSIBLE
Liver disease, or AST/ALT > 120	Hepatotoxicity 0.6 · Elevated LFTs 0.3 · Cholestasis 0.1	60%
CKD, or creatinine > 2.0	Nephrotoxicity 0.5 · AKI 0.3 · Electrolyte imbalance 0.2	50%
Cardiac disease, or QT-prolonging drug	Arrhythmia 0.5 · QT prolongation 0.3 · Bradycardia 0.2	50%
Warfarin with CYP2C9 poor metaboliser	Bleeding 0.7 · Bruising 0.3	70%
HLA risk allele present	SJS/TEN 0.4 · Hypersensitivity 0.3 · Rash 0.3	40%

Consider a patient with chronic kidney disease. The generator will record nephrotoxicity half the time, acute kidney injury three times in ten, and an electrolyte disturbance twice in ten. A model that has perfectly learned this rule still answers "nephrotoxicity" and is wrong half the time. **Weighting each branch by how often it occurs gives an overall maximum of 58.27%.**

Why this matters beyond this project

Every prediction task has a ceiling below 100%, because outcomes are not fully determined by the information available. Here the ceiling can be calculated exactly, since the data was generated by a known rule. With real patients it cannot be calculated — but it still exists. A tool advertising 99% accuracy on a genuinely uncertain clinical outcome is telling you something is wrong with the evaluation, not that the model is exceptional.

6.4 What changed, and why it changed so much

Multiclass accuracy before and after

CONFIGURATION	ACCURACY	VS BASELINE
Previous model — 200 records, 60 classes	15.50%	−74 pts
Retrained on 50,000 records, 20 classes	42.93%	+20.6 pts
Also removing <code>class_weight="balanced"</code>	52.72%	+30.4 pts

Almost all of the gain came from one thing: **the model had been trained on 200 records spanning 60 outcome classes** — about three examples per class. The generator defaults to 200,000 records and had been run with 200. Regenerating the cohort at 50,000 records collapsed the class count to 20 and raised the rarest class from 3 examples to 265.

The second change was removing `class_weight="balanced"`, which instructs the model to treat rare classes as equally important. That is often the right choice for rare clinical outcomes. Here it cost 10 points of accuracy while leaving balanced accuracy essentially unchanged (29.9% against 28.8%) — it was paying a real price for nothing, so it was removed.

6.5 Per-class performance

Overall accuracy hides wide variation. The model handles the mechanistically distinct reactions well and the non-specific ones poorly:

Held-out test set, selected classes

ADR CLASS	PRECISION	RECALL	F1	SUPPORT
Bleeding / Haemorrhage	0.691	0.889	0.778	126
No ADR	0.595	0.999	0.746	2,229
Hepatotoxicity	0.591	0.875	0.706	546
Cardiovascular Event (Arrhythmia)	0.503	0.935	0.654	1,782
Nephrotoxicity	0.485	0.809	0.606	698
Severe Cutaneous Reaction (SJS/TEN)	0.474	0.635	0.543	85
Acute Kidney Injury	0.286	0.145	0.192	441
QT Prolongation	0.294	0.055	0.093	1,046
Gastrointestinal (Nausea/Vomiting)	0.000	0.000	0.000	486
Headache / Dizziness	0.000	0.000	0.000	195

Precision — of those flagged as this reaction, the proportion that truly were. **Recall** (sensitivity) — of those that truly were, the proportion detected. **Support** — how many such cases existed in the test set.

The pattern is clinically legible. Reactions with a specific mechanistic trigger in the data — bleeding on warfarin with a CYP2C9 poor-metaboliser genotype, hepatotoxicity with deranged LFTs — are identified well. Non-specific reactions such as nausea and headache are never identified at all, because the generator emits them from an undirected branch with no distinguishing features. **The model cannot separate what the data does not separate.**

Two competing readings of the same row

Note "No ADR": recall 0.999 but precision 0.595. It catches virtually every genuine no-reaction case, but four in ten of the patients it clears will in fact have a reaction. In a clinical tool that direction of error — false reassurance — is the dangerous one, and it is invisible in the accuracy figure.

6.6 Claims that were removed

Two fabricated metrics used to be written into every report

`scripts/model_trainer.py` previously emitted these as fixed text, regardless of measured results — a claim of "> 92% recall" for pharmacogenomic hypersensitivities, and an ablation reporting "weighted F1 0.871 → 0.774". Neither was supported by any artifact in the repository; the real weighted F1 has never exceeded 0.41. Both have been deleted, and the report generator now emits only figures produced by the run itself.

This is recorded rather than quietly fixed because recognising the pattern is the transferable skill: a stated metric is only as good as the experiment behind it, and a number that cannot be traced to a reproducible run should not be believed.

6.7 What is still missing

The figures above are honest measurements on synthetic data. They say nothing about real patients. Reaching clinical credibility would still require:

1. **Real data** with linked outcomes, under appropriate governance.
2. **Calibration assessment** — a stated 70% risk should correspond to the event occurring in about 70% of such patients. Untested here.
3. **External validation** on a different institution's population.
4. **Decision-threshold analysis** — the 25/50/75% tier cut-points are round numbers, not values derived to achieve a target sensitivity.
5. **Prospective evaluation** of whether using the tool changes outcomes.

7 Layer 3 — The AI narrative

The third layer uses a large language model to convert the numerical output into clinical prose: an interpretation, monitoring suggestions, and a mitigation plan.

Provider	NVIDIA NIM (OpenAI-compatible endpoint)
Primary model	<code>nvidia/nemotron-3-super-120b-a12b</code>
Fallback model	<code>nvidia/nemotron-3-ultra-550b-a55b</code>
Final fallback	Deterministic rule-based text

Graceful degradation

Language model endpoints fail routinely — they return "temporarily overloaded", or take too long. The system tries the primary model, retries once if it reports itself overloaded, falls through to the secondary model, and finally emits fixed rule-based text. A total time budget bounds the whole sequence so a request cannot hang.

Every response carries an `ai_generated` flag, so it is always possible to tell whether the text was written by the model or by the deterministic fallback.

Fluency is not accuracy

A language model produces text that reads as though written by a confident clinician, whether or not the content is correct. It has no access to the patient beyond the fields supplied, and it cannot verify the risk score it is asked to explain. The risk score it is given now carries real signal (Section 6), but it is derived from synthetic training data, and the narrative will read with exactly the same confidence whether the underlying number is sound or not. Every AI-generated statement requires independent verification.

8 Safety, limitations and governance

Not suitable for clinical use

This system must not inform decisions about any real patient. The limitations below are not caveats around a working tool; several are fundamental.

Model limitations

- **Trained entirely on synthetic data** — no real patient contributed, so no figure in Section 6 describes real-world performance. This remains the fundamental limitation, and no amount of tuning addresses it.
- **Never externally validated** — no prospective or independent evaluation.
- **Uncalibrated** — the stated percentages have no verified relationship to real event frequencies.
- **Non-specific reactions are never identified** — nausea, headache and similar classes score zero, because the data contains nothing to distinguish them.
- **False reassurance is the dominant error mode** — "No ADR" has precision 0.595, so a substantial share of patients it clears would in fact react.
- **Risk tiers are arbitrary** — 25/50/75% are round numbers, not thresholds derived to achieve a target sensitivity.

System limitations

- **No authentication** — every endpoint is publicly accessible.
- **No database** — assessments are held in memory and lost on restart.
- **No audit trail** — a clinical requirement that is entirely absent.
- **No regulatory clearance** — a tool of this type would be a regulated medical device in most jurisdictions.

What the project does demonstrate

It is a working example of a full clinical decision support architecture: validated data capture, a served ML model, deterministic rule engines, language model integration with sensible failure handling, and a documented deployment. The model now discriminates genuinely on the task it was given, and Section 6.4 is itself instructive — the largest single gain came not from a better algorithm but from noticing the training set had 200 rows instead of 200,000. As a teaching artifact for how such systems are built, and how to evaluate the claims they make, it is useful. As a clinical instrument, it is not.

What to take away

- Always ask what baseline an accuracy figure beats.
- Ask whether the training data is real, and whether validation was external.
- For rare outcomes, demand per-class recall, not overall accuracy.
- Check whether a quoted metric can be traced to a reproducible run.
- Treat fluent AI prose as a presentation layer, never as evidence.

9 Glossary

TERM	IN CLINICAL LANGUAGE
Accuracy	Proportion of all predictions that were correct. Misleading when one outcome dominates.
Precision	Positive predictive value — of those flagged, how many truly had the outcome.
Recall / sensitivity	Of those who truly had the outcome, how many were detected.
F1 score	Harmonic mean of precision and recall; a single number balancing the two.
Support	Number of true cases of a class in the test set — the denominator behind the percentages.
Baseline	The score achievable by a trivial rule. A model must beat it to be worth anything.
Class imbalance	One outcome vastly outnumbering others — as with any rare disease.
Calibration	Whether a stated 70% risk corresponds to the event occurring in 70% of such patients.
Training / test split	Holding data back to test on, so performance is not measured on cases already seen.
External validation	Testing on a different population or institution — the standard for real clinical claims.
ROC-AUC	Given one patient with the outcome and one without, the chance the model ranks them correctly. 0.5 is a coin toss, 1.0 is perfect.
Balanced accuracy	Mean recall across all classes, so a rare class counts as much as a common one.
Bayes-optimal ceiling	The best score any model could achieve, set by irreducible randomness in the outcome.
Class weighting	Telling the model to treat rare classes as equally important, trading overall accuracy for rare-class detection.
Gradient boosting	Many simple decision trees in sequence, each correcting its predecessors' errors.
Feature	One input variable, e.g. eGFR or age.
Hyperparameter	A setting chosen before training, e.g. how many trees to build.
Synthetic data	Computer-generated records, not from real patients.
Large language model	A text-generating model. Fluent by construction; not a source of verified fact.
Pharmacogenomics	How inherited genetic variation alters drug response.
CYP enzymes	Cytochrome P450 — the main family metabolising most drugs.
Poor metaboliser	Reduced enzyme activity, so the drug accumulates at standard doses.
HLA allele	Immune system variant; specific alleles predict severe hypersensitivity to particular drugs.

PhenoRx Technical & Methodological Overview · System version 2.1.0. All performance figures in this document were taken from the repository's own artifacts or measured directly from the shipped model; none are estimated. The model evaluated is `models/adr_model.pkl`.

This document describes a research and teaching prototype. It is not a clinical tool, has no regulatory clearance, and must not be used in the care of any patient.